

José Manuel Robles-Aguilar

M.Sc. IN STATISTICAL PHYSICS · ARTIFICIAL INTELLIGENCE · PROTEIN SCIENCE · BIOPHYSICS

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Education

Cinvestav

M.Sc. IN PHYSICS

- GPA: 3.6

Mexico City, MX

01 Sep 2023 - 16 Dec 2025

Center for Computing Research (CIC)

SEMESTER

- Algorithm design and analysis, and quantum computing courses

Mexico City, MX

01 Sep 2022 - 01 Jan 2023

National Polytechnic Institute (IPN)

BACHELOR DEGREE IN PHYSICS AND MATHEMATICS

- GPA: 3.4

Mexico City, MX

01 Sep 2017 - 01 Jul 2022

Research Experience

Thesis: “AI-driven Prediction of Circular Dichroism Spectra of Proteins”

CINVESTAV - DEPT OF PHYSICS

- Advisors: Dr. Mauricio Demetrio Carbajal-Tinoco, Dr. Damián Jacinto-Méndez
- Implemented deep Bayesian neural networks into the KCD model increasing the accuracy by 40%

Mexico City, MX

Nov 2024 - Nov 2025

AI+Science Summer School

UNIVERSITY OF CHICAGO JOHN W. BOYER CENTER

- Organizer: University of Chicago Data Science Institute

Paris, FR

Jul 2025

Fluorescence Microscopy project: “Brownian-particle motion inside lipid vesicles”

SOFT MATTER LABORATORY - CINVESTAV - DEPT OF PHYSICS

- Advisor: Dr. Damián Jacinto-Méndez

Mexico City, MX

May 2024 - Jul 2024

Proteomics Project: “Radius of Gyration in Alpha-helix and Beta-sheet proteins”

CINVESTAV - DEPT OF PHYSICS

- Advisor: Dr. Mauricio Demetrio Carbajal-Tinoco

Mexico City, MX

Jan 2024 - Feb 2024

International Research School

DURHAM UNIVERSITY

- Organizers: Durham University, Queen Mary University of London, University of Leeds

Durham, UK

Jan 2022

Teaching Assistantship

CENTER FOR COMPUTING RESEARCH (CIC)

- Advisor: Dr. Sun GuoHua

Mexico City, MX

Sep 2021 - Mar 2022

Institutional Incentive Grant for Research Training (BEIFI)

SUPERIOR SCHOOL OF PHYSICS AND MATHEMATICS (ESFM-IPN)

- Advisors: Dr. Gerardo Ortega-Cervantes, Dr. Ana María Paniagua-Mercado and Dr. Norma Sanchez-Salas

Mexico City, MX

2018 - 2022

Poster Presentations

Robles-Aguilar, J. M., Carbajal-Tinoco, M. D., Jacinto-Méndez, D. 2025. *AI-driven Prediction of Circular Dichroism Spectra of Proteins*. 13th Symposium on Structural Proteomics, Milan, Italy.

Robles-Aguilar, J. M., Carbajal-Tinoco, M. D., Jacinto-Méndez, D. 2025. *The knowledge-based Circular Dichroism (KCD) method*. Physics and Life Days Conference, Paris, France.

Skills

Deep Learning, Machine Learning, Ubuntu, Python, PyTorch, Structural Biology, Proteomics, Biophysics, Bioinformatics, Data Analysis, Statistics, Statistical Physics, Quantum Computing, Fluorescence Microscopy, XRD, AFM, EM, UV-Vis, Raman.

References

- **Dr. Mauricio Demetrio Carbajal-Tinoco**, Cinvestav, Department of Physics. **email:** mauricio.carbajal@cinvestav.mx
- **Dr. Damián Jacinto-Méndez**, Cinvestav, Department of Physics. **email:** damian.jacinto@cinvestav.mx
- **Dr. José Matías Alvarado-Mentado**, Cinvestav, Dep of Computer Science. **email:** matias@cs.cinvestav.mx